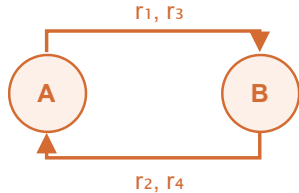


(a) Coarse quotient

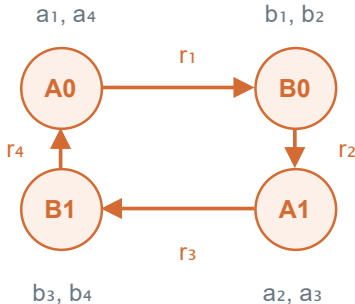


Concrete witness W_1

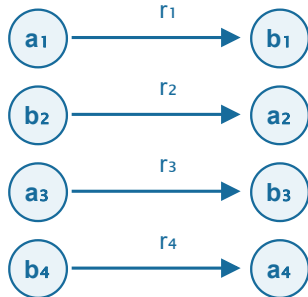
$a_1 \rightarrow b_1 \equiv b_2 \rightarrow a_2 \equiv a_1$

cheap breaks $a_2 \equiv a_1$

(b) Cheap feature: cost 1



(c) Separating feature: cost 2



Every occurrence has a distinct value

Round 1: cover $W_1 \rightarrow$ cheap (cost 1), but the full quotient still cycles

Round 2: add $W_2 \rightarrow$ replace cheap with repair (cost 2); full quotient is a DAG